

# BACKTEST & ATTRIBUTION

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01 THE ENGINE, NET

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YOU BELIEVE IT

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04 WHO EARNED IT —  
THEN WRITE IT

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Your books meet the engine: priced net, checked before believed, then taken apart until the idiosyncratic return has a name.

# WHO BROUGHT THEM.

Every weight file, and the blueprint's commit hash.

01

**YOUR CONTROL**

Same rule, one ingredient out.

Portfolio/

02

**TWO LEADS, READ**

DeMiguel, Garlappi & Uppal first.

Bibliotheca/

03

**YOUR RECONCILIATION**

Every difference named.

JOURNAL\_1.md

04

**YOUR ALTERNATIVE**

Same names, another sizing: a trial.

TRIALS.csv

**WHY THE HASH** The blueprint predates the result, or it is not a protocol.

# WEIGHTS IN. FINDINGS OUT.

Price every book net, check the run, take it apart, ask who earned the residual, write it down.

|                                                              |                                                       |                                                   |                                                        |                                                                      |                                                                      |                                                                               |                                                               |
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THE RULE Every number today comes from the engine or the library. None from memory.

# THE ENGINE, NET

eight assumptions · one engine · costs on, or no number

# FOUR MORE, PRICED TODAY.

Point in time, survivorship, the lag, the rebalance rule:  
yours since session 04.

**01****EXECUTION  
PRICE**

Fill on VWAP, mark on  
the adjusted close.

**02****TRANSACTION  
COSTS**

Commission per share,  
plus slippage.

**03****CAPACITY**

Spread and impact: can  
your capital trade it?

**04****CAPITAL PATH**

Cash pays commission.  
Commission needs  
cash.

**NET OR NOTHING** On a high-turnover book, costs decide whether the edge survives.

# SIX FIELDS. ONE SIMULATION.

Backtest → Parameters → New Backtest. Then Simulate.

**01****PORTFOLIO**

Your session 04 Lab job.

Portfolio job #N

**02****BENCHMARKS**

At least one. The first is primary.

your JOURNAL\_1 pick

**03****CASH RESERVE (%)**

Default 1. Pays the commission.

2

**04****COMMISSION (¢ PER SHARE)**

Read the unit twice.

0.005

**05****EXECUTION REALISM**

Cost preset, from Curator data.

Spread + impact

**06****DELISTING**

Price data ends early.

Sell at the last available price

**THE TRAP** Labelled ¢, charged in \$: the default 0.05 is five cents a share.

# SAME ENGINE. YOUR BOOKS.

Section 4 of experiment\_1.ipynb: every book, one window, net.

## PASTE THIS INTO YOUR ASSISTANT

1 · Using backtest-engine-runs, in section 4 of experiment\_1.ipynb, price portfolio\_weights.csv, the control and the alternative over one window, net: a 2% cash reserve, a per-share commission, slippage in basis points, BLUEPRINT\_1.md's benchmark.

2 · Then section 8, Verify: every run valued to the end of the window it was asked for.

---

Behind the line: Experiments/backtest\_engine.py writes, runs, reads back and aligns. Results land in Backtest/, never committed.

### 01

#### ONE WINDOW

Every arm: same dates, same costs.

### 02

#### THE BENCHMARK

The blueprint's, not today's.

### 03

#### DAILY\_WEIGHTS

Saved: attribution reads it.

# THE BOOK SUMS TO ONE. WHO PAYS?

Both pass session 04's five checks.

**01****NO CASH, NO  
COMMISSION**

Weights sum to 1:  
nothing left to pay.

Cash error on <date>

**02****SET A RESERVE**

A fraction. The example  
needed 0.02.

cash\_reserve\_percentage

**03****READ THE UNIT**

Named cents, charged  
in dollars a share.

commission\_cents

**04****CHECK IT ONCE**

Commissions over  
shares traded.

orders\_df

**THE CURE** Price one run, divide Total\_commissions by shares traded, journal the rate.

# READ IT BEFORE YOU BELIEVE IT

a short run looks better · one window is an anecdote · the control decides

# THE SHORTER RUN LOOKED BETTER.

One book, run twice. Reserve 0, then a reserve. Engine 0.66.0.

|                         | TRUNCATED   | COMPLETE    |
|-------------------------|-------------|-------------|
| success / error         | True / None | True / None |
| Days valued             | 522         | 1,305       |
| CAGR                    | 23.6%       | 14.2%       |
| A field naming the stub | none        | —           |
| Excel report written    | yes         | yes         |

THE CHECK end\_date and years against the window you asked for. Before any metric.

# EVERY ROW, PRICED TODAY.

One window is one anecdote. Price what the blueprint declared.

**01****RULE AND  
CONTROL**

Same dates, one  
ingredient apart.

Experiment\_1

**02****TWO COST  
ROWS**

The blueprint's  
commission, and a  
realistic one.

headline · realistic

**03****SUB-PERIODS**

The windows the  
blueprint fixed.

2006-11 · 2012-19 · 2020-24

**04****THE BUDGET**

At most 10 of the demo's  
20 backtests.

≤ 10 runs

# IT BEAT THE INDEX. AND FAILED.

Net, 2017-01-03 to 2026-06-01, Backtest Engine 0.66.0. KN600 is the example’s index.

| ARM                       | CAGR   | VOL    | SHARPE | MAX DD  |
|---------------------------|--------|--------|--------|---------|
| The rule                  | 17.87% | 22.18% | 0.8057 | -32.20% |
| The control               | 18.73% | 24.17% | 0.7748 | -41.83% |
| Diagnostic arm            | 19.52% | 22.25% | 0.8773 | -32.36% |
| The rule, realistic costs | 18.36% | 22.17% | 0.8280 | -32.15% |
| KN600                     | 14.63% | 19.01% | 0.7699 | -33.75% |

**KILL SWITCH** Ahead of its control on both measures in one sub-period of three. It trips.

# ATTRIBUTION, THREE CUTS

Every return has a reason. Where is yours coming from?

# THREE CUTS, ONE BOOK.

Each cut asks a different question of the same daily weights.

**01**

## FIRST CUT

Brinson-Fachler: allocation, selection, interaction.

BrinstonFachlerArrowAttribution

**02**

## SECOND LAYER

Factor model: which premia paid, and what is left.

KNFMArrowAttribution

**03**

## THIRD PASS

Brinson-Fachler on the residual.

built in the notebook

**THE LIBRARY'S WAY** Per asset, per date: allocation is a sector bet only if you grouped first.

# EVERY RETURN HAS A REASON.

Exposures per stock, per date: a cross-sectional, Fama-MacBeth model.

**01****MARKET**

The index's own move.

`f_market`**02****STYLE · 5**

Beta, size, value, residual volatility, momentum.

`f_beta ... f_value`**03****INDUSTRY · 11**

The GICS sectors.

`f_<GICS sector>`**04****IDIOSYNCRATIC**

What no factor explains.

`idio_returns`

# WIDEN THE BOOK, OR THE FIRST CUT LIES.

A book of 8 names inside a 788-name index.  
Attribution Analysis 0.2.0.

| FIRST CUT                                   | BOOK ONLY | WIDENED   |
|---------------------------------------------|-----------|-----------|
| Benchmark return, as a share of the index's | 6 . 0%    | 98 . 9%   |
| Active return (alpha column)                | +0 . 8845 | +0 . 1735 |
| Interaction                                 | +0 . 7415 | +0 . 0399 |
| Portfolio return                            | +0 . 9305 | +0 . 9305 |

**THE RULE** Every constituent in the weight file: zero where not held, each with a price series.

# TAKE IT APART, IN CODE.

Section 5 of `experiment_1.ipynb`: two layers, every arm.

## PASTE THIS INTO YOUR ASSISTANT

- 1 · Using `attribution-analysis-runs`, in section 5 of `experiment_1.ipynb`, read each run's `Daily_Weights`, widen it to every benchmark constituent at zero, then run Brinson-Fachler and the factor model.
- 2 · Beside every number: the coverage line, the factor files, the common window.

---

Behind the line: `Experiments/attribution_analysis.py` shapes the four inputs. `main()` writes no files: the tables come from the objects.

**01****DAILY, NOT REBALANCE**

The library rejects the weight file.

**02****SAME PRICE BASIS**

The adjusted close the engine marked on.

**03****COVERAGE, ALWAYS**

No factor split without it.

# MOSTLY MARKET. THEN MOMENTUM.

The rule, from its first trade on 2017-01-04 to 2026-06-01: 160.02 points of excess. Attribution Analysis 0.2.0.

| FACTOR MODEL                      | POINTS             |
|-----------------------------------|--------------------|
| Market                            | 87.39              |
| Momentum                          | 15.44              |
| Beta · Size · Residual volatility | 7.60 · 6.86 · 5.82 |
| Value                             | 0.71               |
| Eleven sectors                    | 0.00 each          |
| Idiosyncratic                     | 36.19              |

FIRST CUT    Active return +52.14 = allocation +0.11, selection +12.26, interaction +39.78.

# WHO EARNED IT — THEN WRITE IT

one thing removed, each time · the model's blind spot · the findings · then stop

# ONE THING REMOVED, EACH TIME.

Each is a weight file the engine prices: same window, same costs.

## PASTE THIS INTO YOUR ASSISTANT

Using alpha-decomposition, in section 6 of experiment\_1.ipynb, price the arms BLUEPRINT\_1.md names: equal weight within each date, up to 3 random draws from the eligible pool at the same sizes, and entries shifted one day. Publish K.

**01****SIZING**

Same names, equal weight.

**02****SELECTION**

Up to 3 random books, same sizes.

**03****TIMING**

Entries shifted one day.

**04****THE FILTER**

Signal off: your control.

# THE MODEL CANNOT SEE A TREND.

Factor momentum ranks names against each other. A trend rule compares a name with its own past.

**01**

## RELATIVE

Momentum: this name against its peers.

**02**

## ABSOLUTE

Trend, breakout, regime: against itself.

**03**

## THE TEST

Same book, signal off. Price both.

# THE CROSS COSTS IDIOSYNCRATIC RETURN.

Idiosyncratic points, factor model,  
same window. Attribution Analysis  
0.2.0.

| ARM                                | IDIOSYNCRATIC POINTS |
|------------------------------------|----------------------|
| The rule                           | 36.19                |
| The control · golden cross off     | 40.99                |
| Diagnostic arm                     | 50.02                |
| Random books · 5 draws, same sizes | -11.78 to 35.31      |
| Random books · mean                | 22.4                 |

**READING** The liquidity ranking beats random names. The cross in the name subtracts.

# WRITE IT SO IT CAN BE CHALLENGED.

FINDINGS\_1.md, from the notebook's outputs.  
Session 06 reads nothing else.

## PASTE THIS INTO YOUR ASSISTANT

Using experiment-lifecycle, fill FINDINGS\_1.md from experiment\_1.ipynb's outputs: the book and its control, net, both cost rows, both attribution layers with coverage and window, the arms, every excluded run by name, the trial count. Quote every number from engine\_summary.csv or run\_manifest.csv; compute none.

Behind the line: the findings follow the template's sections, and RESULTS.md is compiled from them, never written by hand.

### 01

#### THE BOOK, PRICED

Every row on one window, net.

### 02

#### BOTH LAYERS

Each number with its coverage.

### 03

#### THE COUNT

Every variant you ranked.

# STOP HERE, ON PURPOSE.

## ON DISK WHEN YOU LEAVE

---

- Backtest/ — every row priced net, one window
- Section 8, Verify — every run valued to the end
- Attribution/ — two layers, coverage logged
- run\_manifest.csv — every run, with its window and costs
- FINDINGS\_1.md — every number, quoted from a run
- RESULTS.md — compiled from the findings
- A JOURNAL\_1.md entry — each excluded run, by name

Left: on disk before you leave. Right: not done, on purpose.

## NOT TODAY

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- The arms and the holdout — homework, in the window
- The third pass — homework, if you can
- The challenge and the gate — session 06
- Experiment 2 — a new blueprint first

## WHY WE STOP

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A rule changed after the result is a new experiment, with a new blueprint.

# TO DO.

Commit the method and the documents, never  
Backtest/.

**01****YOUR BOOK,  
NET**

Priced by the engine,  
costs on.

FINDINGS\_1.md

**02****YOUR CONTROL**

One ingredient out,  
same dates.

Portfolio/

**03****THE  
HOMEWORK  
RUNS**

A one-day shift, up to 3  
random books, the  
holdout once.

TRIALS.csv

**04****YOUR COUNT**

Every variant you  
ranked.

RESULTS.md

**IF YOU CAN** The third pass: Brinson-Fachler on the residual. Almost nobody does; the gate asks.

# BACK TO THE PILLARS.

The coding, markets and math sessions behind today. Go back to one when you could not do its part with the AI off.

## CODING · CODING FOUNDATIONS

- CF S10 Architecture of a KaxaNuk Library  
guarded imports of licensed libraries
- CF S11 House Rules and Code Review  
reviewing the agent’s draft against the checklist

## MARKETS · FINANCIAL MARKETS

- FM S07 Liquidity, Trading Costs and Capacity  
costs with units, commission on unadjusted shares
- FM S12 Capstone: Performance, Attribution and Track Records  
engine metrics and Brinson-Fachler
- FM S02 Prices, Total Returns and Time Value  
total returns and the unadjusted price
- FM S10 CAPM, Beta and the Market  
beta
- FM S11 Factor Models: Risk Factors and Return Signals  
the factor layer
- FM S09 Risk, Diversification and the Long-Only Book  
setting the cash reserve before the first run

## MATH · MATHEMATICAL FINANCE

- MF S01 Return Arithmetic  
annualisation, Sharpe, the CAGR of a short window
- MF S02 Distributions, Tails and Drawdowns  
tails, VaR, CVaR
- MF S06 Regression and Its Failures  
regression, for the Ken French diagnostic
- MF S08 Cross-Sectional Signals: IC, Breadth and Fama-MacBeth  
reading a factor layer

# THE READING LIST.

Six leads. Read two; start with Brinson & Fachler: the first cut, from its source.

## THE ATTRIBUTION · WHERE IT CAME FROM

### FAMA & MACBETH

**1973** Risk, Return, and Equilibrium: Empirical Tests  
→ the factor model

### BRINSON & FACHLER

**1985** Measuring Non-U.S. Equity Portfolio Performance  
→ the first cut

### BRINSON, HOOD & BEEBOWER

**1986** Determinants of Portfolio Performance

## THE FACTORS AND THE COSTS · WHAT ELSE IT COULD BE

### FAMA & FRENCH

**1993** Common Risk Factors in the Returns on Stocks and Bonds

### CARHART

**1997** On Persistence in Mutual Fund Performance  
→ f\_momentum

### NOVY-MARX & VELIKOV

**2016** A Taxonomy of Anomalies and Their Trading Costs  
→ net, or nothing

**ALSO READ** The Backtest Engine and Attribution Analysis methodology docs; Paleologo (2025), ‘Ex-Post Performance Attribution’.

# NOW TAKE IT APART.

A curve says what happened. Attribution  
says why. The arms say who.

## THE ENGINE

Backtest/  
Net, verified, one window.

## THE LIBRARY

Attribution/  
Three cuts, coverage stated.

## BRING

FINDINGS\_1.md, your trial count.

**QUESTIONS** [research@kaxanuk.mx](mailto:research@kaxanuk.mx)

NEXT

## SESSION 06

TOPIC

## FINAL STRATEGY PREP & PRESENTATION

YOU LEAVE WITH

**THE CHALLENGE, N, A  
DEFLATED SHARPE, THE  
GATE VERDICT, A  
REHEARSED TALK.**

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